



Armada Kalamunda Group

Agitation and Arousal: Pharmacological Management Guideline

1. Purpose

To guide clinicians in the pharmacological management of agitation and arousal in mental health inpatient units (ages 18 years and above), the associated monitoring and the correct use of the relevant prescribing chart (*Agitation and Arousal PRN Medication Chart AKMR170.2*).

2. Applicability

This guideline applies to medical, pharmacy and nursing staff within Armada Mental Health Services inpatient wards.

3. Aboriginal Health impact statement and declaration policy statement

The policy author has declared that there are no perceived health impacts on Aboriginal people ;in this policy document (ISD Record 4326)

4. Guidelines

4.1 Principles of safe prescribing

- Non-pharmacological interventions should be used as the first line option, where possible, to de-escalate the patient.
- Treatment options are determined by patient's age. Refer to [Appendix 1](#) for 18 - 65 YEARS and [Appendix 2](#) for OVER 65 YEARS.
- Where the listed treatment options are not suitable, consultant psychiatrist input is required to prescribe alternatives. Prescribing rationale must be documented in the medical record.
- Medications for managing acute agitation and arousal must be prescribed on AKMR170.2.
- The aim of pharmacological management is to reduce the patient's agitation and arousal score to 1 and not to sedate the patient.
- Oral route is preferred over parenteral administration.
- Orally disintegrating tablets, wafers, liquids or syrups are preferred to tablet to aid compliance and reduce the risk of diversion. Please specify the preparation on the drug chart particularly if using liquid or syrups.
- Consideration should be given to any known pre-existing medical illnesses, drug allergies, known legal and illegal substances recently taken (alcohol, amphetamines etc.) and any regularly prescribed oral/depot medication, as this may impact on dose requirements and potential side effects.

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- An exact dose must be specified, not a dosage range.
- Each medication order must be for ONE route only.
- The 'Max dose/24 hr' prompt on the *AKMR170.2* chart indicates the total amount of the medication, including both oral and IM formulations, which may be administered for PRN doses only for that medication. The maximum dose allowed within a 24-hour time period does not relate to the calendar day. To determine whether a patient can have a subsequent dose, calculate the cumulative dose of the PRN medication received in the preceding 24 hours, and ensure a subsequent dose will not exceed the maximum specified on the PRN order.
- The prescriber should ensure that the maximum recommended daily dose of a prescribed medication on the PRN order is not exceeded by taking into account any regular doses of that medication prescribed in the 'regular medicines' section of the *WA Hospital Medication Chart*.
- Antipsychotic polypharmacy (concurrent use of two or more antipsychotics, including those prescribed regularly) and high dose prescribing (doses in excess of the licensed maximum daily dose as outlined in the Australian Product Information leaflet) carries risks, therefore the use of antipsychotics for the management of agitation and arousal should be monitored as per the *AKG Antipsychotic Polypharmacy and High Dose Antipsychotic Prescribing procedure*.
- Unless contraindicated, benztropine should be prescribed in the 'when required' (PRN) section of the *WA Hospital Medication Chart (AKMR170B or AKMR171)* when typical antipsychotics are prescribed especially haloperidol. The usual dose of benztropine is PO or IM 1mg – 2mg TDS PRN.
- Patients who may be more sensitive to the effects of psychotropics, such as those who are neuroleptic naïve, dehydrated or underweight, under 24 years, older adults, intoxicated with drugs and alcohol, pregnant, epileptic, or have hepatic or renal impairment, may require lower doses and more frequent observation and monitoring using *Adult Observation and Response Chart AKMR140.3*.
 - Consideration is required when prescribing IMI doses to these persons and a lower dose is likely required given the increased risk of oversedation.
- Baseline ECG monitoring is strongly recommended before prescribing and administering parenteral antipsychotics. ECG monitoring is recommended for all patients receiving haloperidol in any formulation based on clinical need and presentation. If it is not possible to perform an ECG, the reason must be clearly documented in the patient record.
- The rationale for any changes or deviations from this guideline should be clearly documented in the medical notes.
- The treating team or out of hours duty doctor should be notified if an IM injection has been given. A doctor should aim to be at the scene within 30 minutes, where required. Medical support must be available in case of adverse reactions, over sedation (RR< 10 breaths per minute) or the need to administer flumazenil (to reverse sedation, drowsiness). Please contact the DMO or Medical RMO (if DMO not on-shift) or initiate code blue.
 - Resuscitation equipment and medication, including flumazenil, must be available and easily accessible and staff should be familiar with their use. The equipment must be maintained and checked every week.
 - Flumazenil should be used with caution in patients known to be longstanding benzodiazepine users as they may experience symptoms of acute withdrawal or seizures. Avoid using flumazenil in patients who are using benzodiazepines to

control potentially life-threatening conditions e.g. *unstable intracranial pressure* or *status epilepticus*.

4.2 Agitation and Arousal Scores

- Agitation and arousal scores must be documented on the Agitation and Arousal PRN Medication Chart.

Table 1: Agitation and arousal rating scale and associated management

Score	Patient's state	Management
0	Asleep or unconscious	Review prescribed doses and conduct required monitoring.
1	Settled, minimal agitation	Optimal state; PRN medication not required.
2	Mildly aroused, pacing, still willing to talk reasonably	- Attempt to verbally de – escalate the patient, as appropriate - Offer oral (PO) treatment relevant to patient's current rating. - Intramuscular (IM) option only if patient refuses oral treatment.
3	Moderately aroused, agitated, becoming more vocal and unreasonable or hostile	
4	Highly aroused and possibly distressed or fearful, possibly noisy but not violent	
5	Highly aroused and violent toward self, others and/or property	

- Agitation and arousal scores must be documented:
 - before each PRN dose, and
 - reviewed and documented after:
 - 60 and 90 minutes for all oral doses,
 - 30 and 60 minutes for all IM doses.

4.3 Repeating Doses

- Where additional doses are required, administer at the frequency indicated until a score of 1 or maximum daily dosage is achieved:
 - Oral: May be repeated hourly
 - IM: May be repeated after 30 minutes **except** olanzapine IM and clonazepam IM.
 - Olanzapine IM may be repeated after 2 hours and a third dose 6 hours after the first dose if required. Total of 3 doses or 30mg maximum in 24 hours (20mg maximum in 24 hours for older adults over 65 years) whichever occurs first.
 - Clonazepam IM may be repeated after 3 hours, maximum 4mg in 24 hours (higher doses may be used in exceptional circumstances after discussion with consultant psychiatrist).
- Dosing frequencies outside the recommended guidelines require consultant psychiatrist approval.

- If the first dose of oral medication is inadequate, preferably repeat oral medication in STEP 2. If the second dose of oral medication fails to produce an adequate response after 60 minutes, move to STEP 3. STEP 4 should only be used if oral medication is refused/risk assessed as high. Please seek medical opinion if limited or no response after two interventions.
- Seek the advice of a consultant psychiatrist if not achieving a score of 1 at maximum doses recommended.

4.4 Required Monitoring

- Following the administration of medication for agitation and arousal
 - Physiological observations must be taken to allow monitoring of any physical deterioration. The observations must include:
 - Pulse
 - Blood pressure
 - Temperature
 - Oxygen saturation
 - Consciousness levels
 - Respiratory rate
 - Observe for adverse effects: monitor for signs of extrapyramidal side effects (EPSE) (and in particular, laryngeal dystonia), seizures, neuroleptic malignant syndrome, following the administration of antipsychotics.
 - Observe efficacy of treatment to avoid escalation of further disturbed behaviour
- Frequency of monitoring:

Table 2 Frequency of physiological observations post PO and IM medication

Post PO medication	Monitor the patient after ONE hour. Further monitoring beyond 1 hour should be considered if deemed clinically appropriate.
Post IM medication	1st hour: every 15 minutes Thereafter: every 30 minutes until patient is ambulating or minimum 4 hours post IM dose.

- All physiological observations must be recorded on the *Adult Observation and Response Chart (AKMR140.3)*. Scores should be monitored and responses as per instructions on the *Adult Deterioration Detection System (ADDS) score chart*.
- Seek urgent medical advice or initiate code blue protocol if the patient is unconscious or if there is a sudden deterioration in any of the basic observations of temperature, pulse, BP or oxygen saturation.
- If all physiological observations cannot be carried out, or the patient refuses, non-contact observations must still be completed. Respiratory rate and level of consciousness using the “*alert, voice, pain, unresponsive*” (AVPU) system must be recorded. This would include patients held in seclusion. Nursing staff must annotate ® for refused for other vitals on AKMR140.3 and this detail must be documented in the medical record. Observations should be re-offered or re-attempted as the situation dictates.

- If the patient is asleep, the continuous use of pulse oximetry to measure oxygen saturation is recommended until patient is ambulatory.
- ECG to assess for prolongation of QTc interval and haematological monitoring should be performed at the earliest opportunity when parenteral antipsychotics are given. Hypokalaemia, stress, and agitation place the patient at risk of cardiac arrhythmias.

4.5 Medication review

- The agitation and arousal treatment regimen should be reviewed in conjunction with the patient's regularly prescribed medications. Frequent doses of PRN medications may indicate a need to modify the patient's regular regimen.
- The treating team medical officer must review the AKMR170.2 chart every 24 hours if any of the steps on the chart are utilised during weekdays. During out of hours, weekends and public holidays the charts should be reviewed when requested by a nurse or if 2 or more steps had to be utilised to reduce arousal. This review must be documented in the medical record.

5. Roles and responsibilities

Role	Responsibility
HoD Psychiatry	Has oversight regarding provision of training and application of this policy
Nurse Unit Manager	Has oversight regarding provision of training to nursing staff
Medical officer	To ensure prescribing complies with this document (and rationale is articulated when it does not), management plans are formulated and reviewed, with clear documentation of these processes in the medical record.
Pharmacist	To review the medication charts and address any concerns with the prescriber, including escalation to the team consultant psychiatrist, or to the HOD Psychiatry when required.
Nurse	To utilise PRN usage in accordance with this guideline ensuring that the required observations are checked and recorded on AKMR140.3, complete notifications as outlined and to escalate any concerns as appropriate.

6. Legislative context

The following documents are required to give effect to this policy:

- *WA Mental Health Act 2014*
- *WA Medicines and Poisons Act 2014*
- *WA Medicines and Poisons Regulations 2016*
- Health Department of Western Australia (HDWA) [Medication Chart Policy MP 0078/18](#)
- HDWA [Medication Review Policy MP 0104/19](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0086/18](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Matching EMHS: 21](#)

- EMHS [Consent Policy EMHS: 5](#)

7. Supporting information

The following documents support the implementation of this policy:

- [Antipsychotic Polypharmacy and High Dose Antipsychotic Prescribing Procedure MEN-PRO-0499-19](#)
- [Medication Management Procedure AKG-PRO-0255-18](#)
- [Clinical Handover Procedure AKG-PRO-0521-19](#)
- [Identification of Patients in the Armadale Mental Health Service Procedure MEN-PRO-0203-14](#)
- [Recognition and Response to Acute Deterioration Policy AKG-POL-0388-14](#)

8. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated through regular audits of prescribing performed under the guidance of the Head of Psychiatry (or delegate), under the governance of the Mental Health Safety and Quality Committee.

9. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.3 Partnering with consumers
- Standard 5 – Action 5.33 Predicting, preventing and managing aggression and violence
- Standard 5 – Action 5.34
- Standard 8 – Criterion 8.7: Staff are appropriately trained, developed and supported to safely perform the duties required of them.

10. References

1. Behavioural emergencies; eTG Complete, Therapeutic Guidelines.
2. NICE (2015) Violence and aggression: short term management in mental health, health and community settings

11. Document owner

Enquiries relating to this guideline may be directed to:

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Department: Mental Health
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12. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
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V2.0	11/06/2025	11/06/2028	Section 3.1 – added in hyperlinks for appendix 1 and 2 Added consideration is required when prescribing IMI doses to these persons and a lower dose is likely required given the increased risk of oversedation Section 4.0 – added HoD Psychiatry and Nurse Unit Manager and altered nursing and medical responsibility
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13. Authorisation

This guideline has been authorised and issued by the Service Director Mental Health.

Authorised by	Karen Elliott
Authorisation date	30/04/2025
Published date	11/06/2025

Appendix 1: Agitation and arousal pharmacological management for ADULT mental health inpatients from 18 to 64 years old

Step 1 Agitation and arousal rating 2-3 and willing to take oral medications	Step 2 Agitation and arousal rating 3-5 and willing to take oral medications	Step 3 Agitation and arousal rating 3-5 and willing to take oral medications	Step 4 Agitation and arousal rating 3-5 and UNWILLING to take oral medications
Oral benzodiazepine <u>OR</u> Oral antipsychotic	Higher dose of oral Benzodiazepine (compared to Step 1) <u>OR</u> Higher dose of oral Antipsychotic (compared to Step 1)	Combination of oral benzodiazepine <u>AND</u> oral antipsychotic	IM benzodiazepine <u>OR</u> IM antipsychotic <u>OR</u> Combination of IM benzodiazepine <u>and</u> IM antipsychotic

Recommended pharmacological treatment options:

	Oral	Intramuscular
Benzodiazepine	Lorazepam 1mg – 2mg Maximum 10mg in 24 hours	Clonazepam 1mg - 2mg Maximum 4mg in 24 hours <u>OR</u> Midazolam 2.5mg - 10mg Maximum 20mg in 24 hours
Antipsychotic	Quetiapine Immediate Release 25mg – 100mg Maximum 600mg in 24 hours <u>OR</u> Olanzapine 5mg – 10mg Maximum 30mg in 24 hours <u>OR</u> Chlorpromazine 25mg – 100mg Maximum 600mg in 24 hours	Haloperidol 5mg - 10mg Maximum 20mg in 24 hours <u>OR</u> Olanzapine IM 5mg - 10mg Maximum 30mg in 24 hours **Do not combine or administer IM Olanzapine within 1 hour before or after any parenteral benzodiazepine** <u>OR</u> Droperidol IM 5mg - 10mg ¹ Maximum 20mg in 24 hours
Combination benzodiazepine AND antipsychotic	Combinations of above	Clonazepam IM 1mg – 2mg AND Haloperidol IM 5mg - 10mg

Agitation and Arousal: Pharmacological Management Guideline

*Maximum dose refers to total maximum oral and IM dose (including regular)

¹ Warnings on the use of Droperidol (from eTG)

A number of hospitals have withdrawn Droperidol from their formularies and the drug has been deleted from some clinical guidelines following the US Food and Drug Administration (FDA) Black Box warning concerning potential for cardiac complications. However, there is no convincing evidence for a causal relationship between therapeutic Droperidol administration in the present context and life-threatening cardiac events [NB1]. Droperidol is preferred over haloperidol for behavioural emergencies because it is more sedating, has a quicker onset of action, a shorter half-life and is less cardiotoxic than haloperidol.

Kao LW, Kirk MA, Evers SJ, Rosenfeld SH. *Droperidol, QT prolongation, and sudden death: what is the evidence?* *Ann Emerg Med* 2003;41(4):546-58. [URL]

Monitoring:

- Physiological observations: pulse, blood pressure, temperature, oxygen saturation, conscious levels, respiratory rate
 - Oral: At least after ONE hour (further monitoring beyond 1 hour should be considered if deemed clinically appropriate)
 - IM: every 15 minutes for 1 hour, then every 30 minutes until patient is ambulating or minimum 4 hours post IM dose
- Observe for adverse effects: extrapyramidal side effects (EPSE), seizures, neuroleptic malignant syndrome, following the administration of antipsychotics
- If physiological observations cannot be carried out, or the patient refuses, non-contact observations (respiratory rate and level of consciousness using the “alert, voice, pain, unresponsive” (AVPU) system) must be recorded
- ECG prior to and following IM antipsychotics to assess for prolongation of QTc interval

Reviewing response:

- Review and record arousal rating scale on Agitation and Arousal PRN Chart:
 - before and after 60 and 90 minutes for all oral doses
 - before and after 30 and 60 minutes for all IM doses

Repeating doses:

- Oral: hourly
- IM: repeated after 30minutes EXCEPT for olanzapine IM and Clonazepam IM (see details above)
- Doses may be repeated at the frequency indicated until a score of 1 or maximum daily dosage is achieved

Medication choice:

- Haloperidol or droperidol: avoid if antipsychotic naïve, evidence of cardiovascular disease, prolonged QTc, ECG not available, patients with established or suspected Parkinson’s disease or Lewy Body Dementia

Other medications:

- Benzatropine PO or IM 1mg - 2mg TDS PRN should be prescribed in the PRN section of the *WA Hospital Medication Chart* when typical antipsychotics are prescribed
- Flumazenil must be available if benzodiazepines are administered parenterally

Notification:

- The doctor on duty should be notified if an IM injection has been given

Appendix 2: Agitation and arousal pharmacological management for OLDER ADULT mental health inpatients aged 65 and over²

Step 1 Agitation and arousal rating 2-3 and willing to take oral medications	Step 2 Agitation and arousal rating 3-5 and willing to take oral medications	Step 3 Agitation and arousal rating 3-5 and willing to take oral medications	Step 4 Agitation and arousal rating 3-5 and UNWILLING to take oral medications
Oral benzodiazepine <u>OR</u> Oral antipsychotic	Higher dose of oral Benzodiazepine (Compared to Step 1) <u>OR</u> Higher dose of oral Antipsychotic (compared to Step 1)	Combination of oral benzodiazepine AND oral anti psychotic (ONLY on the direction of the treating consultant psychiatrist)	IM Antipsychotic (ONLY on the direction of the treating consultant psychiatrist)

*Maximum dose refers to total maximum oral and IM dose (including regular)

	Oral	Intramuscular
Benzodiazepine	Lorazepam 0.25mg – 1mg Maximum 3mg in 24 hours	N/A
Antipsychotic	Risperidone 0.25mg – 1mg Maximum 4mg in 24 hours OR Quetiapine Immediate Release 12.5mg – 25mg Maximum 400mg in 24 hours OR Olanzapine 2.5mg – 5mg Maximum 20mg in 24 hours	Haloperidol 0.5mg - 1mg Maximum 5mg in 24 hours OR Olanzapine IM 2.5mg - 5mg Maximum 20mg in 24 hours ** Do not combine or administer IM Olanzapine within 1 hour before or after any parenteral benzodiazepine **

Combination benzodiazepine AND antipsychotic	Combinations of above	N/A.
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Recommended pharmacological treatment options

¹ Care must be taken for patients with dementia or possible delirium to screen and treat for possible causes first before considering pharmacological management for agitation or arousal.

Monitoring:

- Physiological observations: pulse, blood pressure, temperature, oxygen saturation, conscious levels, respiratory rate
 - Oral: Every 15 minutes for at least ONE hour (further monitoring beyond 1 hour should be considered if deemed clinically appropriate)
 - IM: every 15 minutes for 1 hour, then every 30 minutes until patient is ambulating or minimum 4 hours post IM dose
- Observe for adverse effects: extrapyramidal side effects (EPSE), seizures, neuroleptic malignant syndrome, following the administration of antipsychotics
- If physiological observations cannot be carried out, or the patient refuses, non-contact observations (respiratory rate and level of consciousness using the "alert, voice, pain, unresponsive" (AVPU) system) must be recorded
- ECG prior to and following IM antipsychotics to assess for prolongation of QTc interval

Reviewing response:

- Review and record arousal rating scale on Agitation and Arousal PRN Chart:
 - before and after 60 and 90 minutes for all oral doses
 - before and after 30 and 60 minutes for all IM doses

Repeating doses:

- Oral: hourly
- IM: repeated after 30minutes EXCEPT for olanzapine IM may be repeated after 2 hours and a third dose 6 hours after the first dose if required (total of 3 doses or 30mg maximum in 24 hours whichever occurs first)
- Doses may be repeated at the frequency indicated until a score of 1 or maximum daily dosage is achieved

Medication choice:

- Haloperidol: avoid if antipsychotic naïve, evidence of cardiovascular disease, prolonged QTc, ECG not available, patients with established or suspected Parkinson's disease or Lewy Body Dementia
- Avoid benzodiazepines if delirium is suspected as they can cause falls, worsen delirium or cause aspiration.
- If a patient has suspected parkinsonism or lewy body dementia, Quetiapine 12.5mg-25mg is the only antipsychotic which should be prescribed.

Other medications:

- Benzatropine PO or IM 1mg – 2mg TDS PRN should be prescribed in the PRN section of the *Hospital Medication Chart* when typical antipsychotics are prescribed
- Flumazenil must be available if benzodiazepines are administered parenterally

Notification:

- The doctor on duty should be notified if an IM injection has been given